BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 8

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Subject Name: Deep Learning

Problem Statement:

Assignment 8 - Implement Pre-trained BERT model for sentiment analysis or text classification on a sample dataset

Code:

```
# 1. Install Dependencies
%pip install -q -U transformers datasets accelerate

# 2. Imports, Environment Setup, and Model Loading
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import torch

from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix

# Set random seed for reproducibility
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# Initialize model and tokenizer
MODEL_NAME = "textattack/bert-base-uncased-SST-2"
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
model = AutoModelForSequenceClassification.from_pretrained(MODEL_NAME).to(device)
model.eval()
print("Device:", device)

# 3. Load and Sample validation dataset
dataset = load_dataset("nyu-mll/glue", "sst2", split="validation")
sample = dataset.shuffle(seed=SEED).select(range(min(500, len(dataset))))
display(pd.DataFrame(sample[:5]))
```

```
# 4. Prediction Batch Helper Function
@torch.no_grad()
def predict_texts(texts, batch_size=32):
    predictions, positive_probabilities = [], []
    for start in range(0, len(texts), batch_size):
        batch = texts[start:start + batch_size]
        encoded = tokenizer(
            batch, padding=True, truncation=True, max_length=128, return_tensors="pt"
        ).to(device)
        probabilities = model(**encoded).logits.softmax(dim=-1)
        predictions.extend(probabilities.argmax(dim=-1).cpu().tolist())
        positive_probabilities.extend(probabilities[:, 1].cpu().tolist())
    return np.array(predictions), np.array(positive_probabilities)

# 5. Evaluate Model Performance
texts = sample["sentence"]
y_true = np.array(sample["label"])
y_pred, positive_probability = predict_texts(texts)

accuracy = accuracy_score(y_true, y_pred)
precision, recall, f1, _ = precision_recall_fscore_support(
    y_true, y_pred, average="binary", zero_division=0
)
print(pd.Series({"accuracy": accuracy, "precision": precision, "recall": recall, "f1": f1}).round(3))

# 6. Confusion Matrix Visualization
cm = confusion_matrix(y_true, y_pred)
sns.heatmap(
    cm, annot=True, fmt="d", cmap="Purples",
    xticklabels=["negative", "positive"], yticklabels=["negative", "positive"]
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("BERT confusion matrix")
plt.show()

# 7. Analyze Top Mistakes and Misclassifications
review = pd.DataFrame({
    "text": texts,
    "actual": y_true,
    "predicted": y_pred,
    "positive_probability": positive_probability,
})
mistakes = review[review.actual != review.predicted].copy()
mistakes["confidence"] = np.where(
    mistakes.predicted == 1, mistakes.positive_probability,
    1 - mistakes.positive_probability,
)
display(mistakes.sort_values("confidence", ascending=False).head(10))

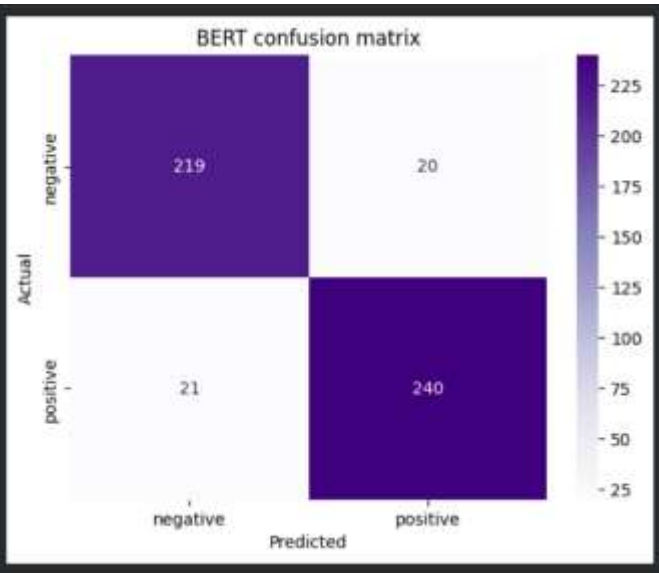
# 8. Custom Sentence Inference
custom_text = ["The story was slow, but the ending was wonderful."]
label, probability = predict_texts(custom_text)
print(custom_text[0], "->", ["negative", "positive"][label[0]], f"(positive probability={probability[0]:.3f})")
```

Output:



	sentence	label	idx
0	it gets onto the screen just about as much of ...	1	726
1	my big fat greek wedding uses stereotypes in a...	1	737
2	for the most part , director anne-sophie birot...	1	51
3	cq 's reflection of artists and the love of ci...	1	482
4	charles ' entertaining film chronicles seinfel...	1	426

```
accuracy     0.918
precision    0.923
recall       0.920
f1           0.921
dtype: float64
```



	text	actual	predicted	positive_probability	confidence
282	the lower your expectations , the more you 'll...	0	1	0.999311	0.999311
184	it 's everything you do n't go to the movies f...	0	1	0.999016	0.999016
278	as unseemly as its title suggests .	1	0	0.001850	0.998150
460	while there 's something intrinsically funny a...	1	0	0.002102	0.997898
366	although huppert 's intensity and focus has a ...	0	1	0.997003	0.997003
327	movie fans , get ready to take off ... the oth...	0	1	0.996455	0.996455
481	every nanosecond of the the new guy reminds yo...	0	1	0.996436	0.996436
183	you wo n't like roger , but you will quickly r...	0	1	0.995712	0.995712
324	harrison 's flowers puts its heart in the righ...	1	0	0.004314	0.995686
244	reign of fire looks as if it was made without ...	1	0	0.004459	0.995541